Dynamic Statistical Arbitrage using Kalman Filtering: A Comparative Pairs-Trading Backtest

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Abstract

This report presents a pairs-trading backtest on the Indian private-bank pair HDFC Bank (`HDFCBANK.NS`) and Kotak Mahindra Bank (`KOTAKBANK.NS`) over the period 2019–2026. Two hedge-ratio estimation methods are compared: a *static* ordinary-least-squares (OLS) hedge ratio combined with an Engle–Granger cointegration test, and a *dynamic* hedge ratio estimated with a Kalman filter that adapts to time-varying co-movement between the two stocks. A rolling z-score of the resulting spread drives a simple mean-reversion entry/exit rule, and both strategies are evaluated on cumulative PnL, Sharpe ratio, maximum drawdown, and trade count. The Kalman filter approach substantially outperforms the static approach on a risk-adjusted basis, achieving a Sharpe ratio of 2.33 versus 0.33 and reducing maximum drawdown from −91.02 to −5.52.

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1 Introduction

Pairs trading is a market-neutral strategy that exploits the tendency of two economically related, cointegrated securities to revert to a stable relationship after temporary divergences. The core idea is to construct a spread

$$\text{spread}_t = y_t - (\beta_t \, x_t + \alpha_t),$$

where β_t (the *hedge ratio*) and α_t (the intercept) describe the linear relationship between the two price series x_t and y_t. When the spread deviates significantly from its historical mean, the strategy shorts the relatively expensive leg and goes long the relatively cheap leg, betting on reversion.

The critical modelling choice is how β_t is estimated:

- **Static hedge ratio.** A single β and α are estimated once via OLS regression over the full sample and held fixed for the entire backtest.
- **Dynamic (Kalman) hedge ratio.** β_t and α_t are treated as a latent state that evolves slowly over time and is re-estimated at every time step using a Kalman filter, allowing the hedge ratio to track structural changes in the relationship between the two stocks.

This report describes the methodology, implementation, and empirical results of both approaches applied to the HDFC Bank / Kotak Mahindra Bank pair.

2 Data

Daily adjusted close prices for `HDFCBANK.NS` and `KOTAKBANK.NS` were downloaded via the `yfinance` API for the period 2019-01-01 to 2026-07-31, yielding 1,875 trading days after alignment and removal of missing observations.

3 Methodology

3.1 Static Strategy

The static hedge ratio is obtained by regressing y_t (Kotak Mahindra) on x_t (HDFC Bank):

$$y_t = \beta x_t + \alpha + \varepsilon_t,$$

estimated once by OLS over the full sample. Before trading, an Engle–Granger cointegration test is run on the price series to confirm that a stationary linear combination exists. The resulting spread is converted to a rolling z-score using a 20-day rolling mean and standard deviation (chosen to avoid full-sample look-ahead bias):

$$z_t = \frac{\text{spread}_t - \mu_{20}(\text{spread})}{\sigma_{20}(\text{spread})}.$$

3.2 Kalman Filter Strategy

The dynamic strategy models the hedge ratio and intercept as a latent state vector $\theta_t = [\beta_t, \alpha_t]^\top$ following a random walk:

$$\theta_t = \theta_{t-1} + w_t, \quad w_t \sim \mathcal{N}(0, Q) \tag{1}$$
$$y_t = H_t \theta_t + v_t, \quad v_t \sim \mathcal{N}(0, R) \tag{2}$$

where $H_t = [x_t,\ 1]$ is the observation matrix, Q is the process (transition) covariance controlling how quickly the hedge ratio is allowed to drift, and R is the observation covariance. The transition covariance is parameterized by a single decay parameter δ:

$$Q = \frac{\delta}{1-\delta} I_2, \qquad \delta = 10^{-5}.$$

At each time step the Kalman filter produces an updated estimate of θ_t using only information available up to time t, so the hedge ratio naturally adapts to structural shifts in the relationship between the two stocks while remaining causal (no look-ahead bias). The same 20-day rolling z-score construction is then applied to the resulting spread.

3.3 Signal Generation

Both strategies use an identical, symmetric entry/exit rule on the z-score:

- Enter short the spread when $z_t > 2.0$.
- Enter long the spread when $z_t < -2.0$.
- Exit (flatten) when $|z_t| < 0.5$.

This produces a position series $p_t \in \{-1, 0, +1\}$.

3.4 PnL Simulation

PnL is computed by marking the position to the change in spread value, net of transaction costs:

$$\text{PnL}_t = p_{t-1} \cdot \Delta\text{spread}_t - c \cdot |\text{spread}_t| \cdot |\Delta p_t|,$$

where $c = 0.001$ (0.1%) is a slippage cost applied to the notional spread value whenever the position changes. Performance is summarized using the annualized Sharpe ratio ($\sqrt{252}\times$ mean daily PnL / standard deviation of daily PnL), maximum drawdown of cumulative PnL, total PnL, and the number of trades (position changes).

4 Results

4.1 Cointegration Test

The Engle–Granger test on the HDFC Bank / Kotak Mahindra Bank price pair returned a p-value of 0.00412, well below the conventional 5% threshold, supporting the presence of a cointegrating relationship and justifying a mean-reversion approach on this pair.

4.2 Performance Summary

Table 1: Backtest performance: static vs. Kalman hedge ratio, HDFC Bank / Kotak Mahindra Bank, 2019–2026.

Metric	Static (OLS)	Kalman Filter
Hedge ratio	0.3044 (fixed)	time-varying (0.39–0.66)
Sharpe ratio	0.33	2.33
Max drawdown	−91.02	−5.52
Total PnL	116.36	135.51
Number of trades	140	152

The static strategy's fixed hedge ratio of 0.3044 (intercept 134.90) is a reasonable full-sample average, but Figure 4 shows that the true relationship between the two stocks drifted considerably over the 2019–2026 window, ranging from roughly 0.39 to 0.66. A hedge ratio fixed at the full-sample average is therefore systematically mis-hedged during large portions of the sample, which is reflected in the static strategy's much larger drawdown and lower Sharpe ratio.

4.3 Static Strategy Plots

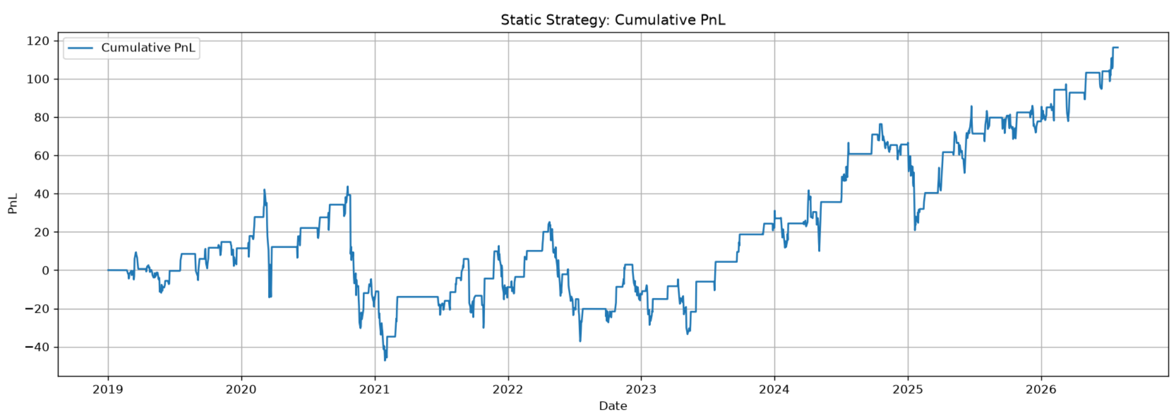

Figure 1: Static strategy: cumulative PnL. The equity curve is choppy and experiences a deep, sustained drawdown through 2020–2021 before recovering and trending upward from 2023 onward.

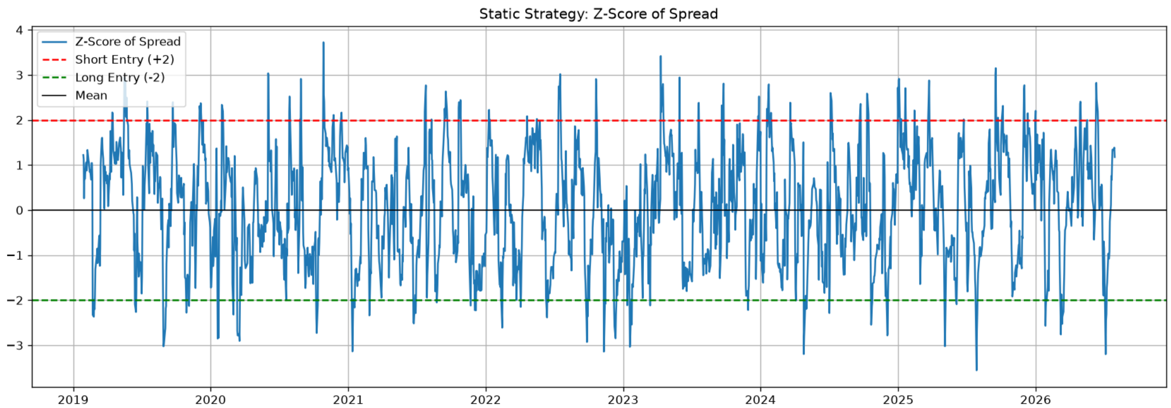

Figure 2: Static strategy: rolling 20-day z-score of the spread, with entry thresholds at ± 2.

4.4 Kalman Filter Strategy Plots

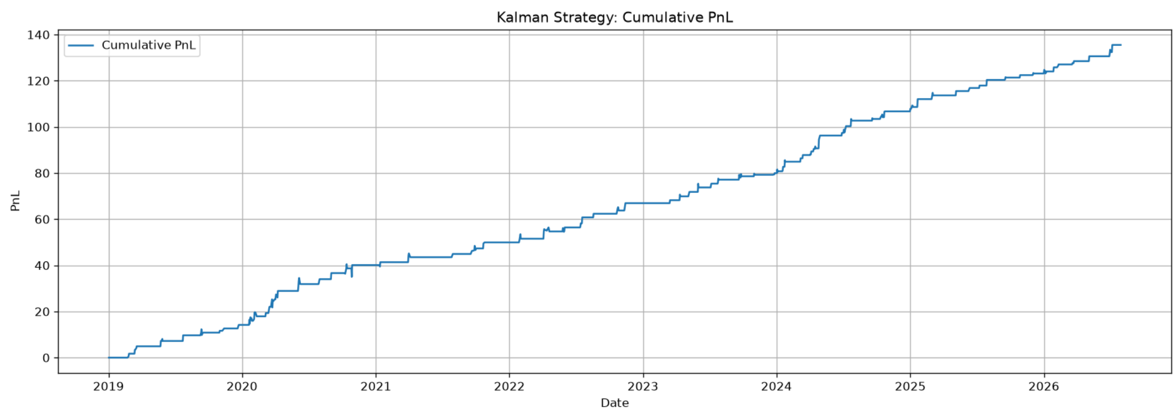

Figure 3: Kalman filter strategy: cumulative PnL. The equity curve is smoother and more monotonic than the static strategy's, with no comparable extended drawdown.

Figure 4: Kalman filter strategy: the time-varying hedge ratio β_t estimated recursively from the data, ranging from approximately 0.39 to 0.66 over the sample period.

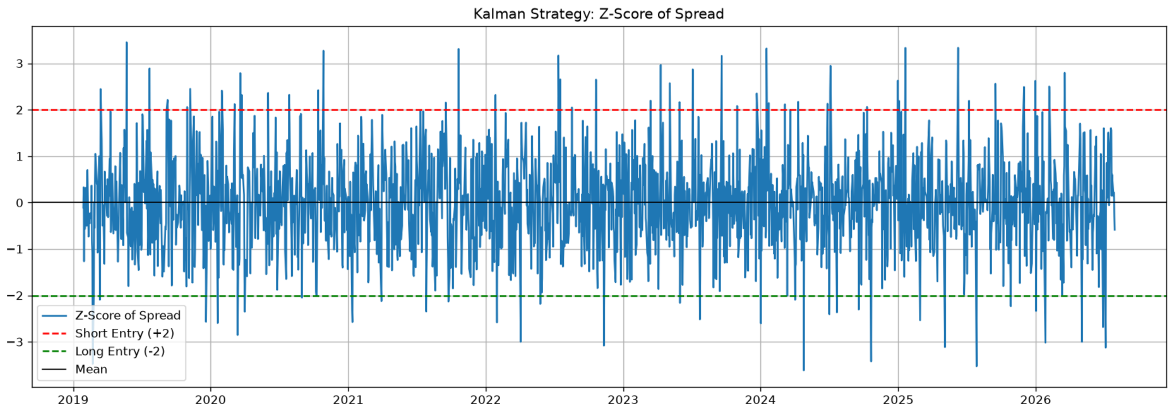

Figure 5: Kalman filter strategy: rolling 20-day z-score of the spread, with entry thresholds at ± 2.

5 Discussion

The results in Table 1 and Figures 1–5 point to a clear conclusion: allowing the hedge ratio to adapt over time materially improves both the risk-adjusted return and the stability of the strategy on this pair.

- **Risk-adjusted return.** The Kalman strategy's Sharpe ratio of 2.33 is roughly seven times that of the static strategy (0.33), despite the two strategies generating similar total PnL (135.51 vs. 116.36). This indicates that the Kalman strategy earns its PnL far more consistently, with fewer and smaller adverse swings.
- **Drawdown.** The static strategy suffers a maximum drawdown of -91.02, nearly the size of its total PnL, meaning that at some point the strategy gave back almost all of its cumulative gains before recovering. The Kalman strategy's maximum drawdown is only -5.52, an order of magnitude smaller, and its cumulative PnL curve (Figure 3) is visibly smoother and more monotonic than the static strategy's (Figure 1).
- **Why the static hedge ratio underperforms.** Figure 4 shows that the true relationship between HDFC Bank and Kotak Mahindra Bank drifted meaningfully over the sample period. A hedge ratio frozen at its full-sample OLS estimate is a poor approximation during periods when the true relationship departs from that average, producing a mis-hedged spread, spurious z-score signals, and larger uncompensated losses – exactly the pattern visible in the static strategy's deep 2020–2021 drawdown.
- **Trade activity.** The Kalman strategy trades slightly more often (152 vs. 140 trades), which is expected since its more responsive spread crosses the entry/exit thresholds somewhat more frequently, but the added transaction costs are more than offset by the improvement in signal quality.

5.1 Limitations

A few caveats apply to this analysis:

- The backtest uses a single pair and a single sample period; results may not generalize to other pairs or regimes.

- Transaction costs are modelled as a simple proportional slippage charge and do not account for market impact, bid-ask spread asymmetries, or borrow costs for the short leg.
- The Kalman filter's process-noise parameter δ was fixed at 10^{-5} without a formal sensitivity or cross-validation analysis; results may be sensitive to this choice.
- Entry/exit thresholds (± 2.0 / 0.5) and the rolling window (20 days) were fixed a priori rather than optimized, so the reported performance is not necessarily the best achievable for either method.

6 Conclusion

This backtest compared a static OLS hedge ratio against a Kalman-filter based dynamic hedge ratio for pairs trading HDFC Bank against Kotak Mahindra Bank over 2019–2026. Both series were confirmed to be cointegrated (Engle–Granger p-value = 0.00412), validating a mean-reversion approach. While both strategies were profitable in absolute terms, the Kalman filter strategy delivered a markedly superior risk-adjusted outcome: a Sharpe ratio of 2.33 versus 0.33, and a maximum drawdown of -5.52 versus -91.02, at a comparable or better level of total PnL (135.51 vs. 116.36) and only a modest increase in trading frequency (152 vs. 140 trades). The key driver of this outperformance is that the true co-movement between the two stocks is not constant over time – the dynamic hedge ratio in Figure 4 ranges from roughly 0.39 to 0.66 across the sample – so a strategy that tracks this relationship in real time hedges more accurately, generates cleaner mean-reversion signals, and avoids the large, sustained drawdowns that plague the static approach. These results support the use of adaptive, state-space hedge-ratio estimation over static regression-based approaches for statistical arbitrage on equity pairs whose relationship is not stable over long horizons. Future work could extend this analysis to a broader universe of pairs, optimize the Kalman filter's noise parameters and the signal thresholds out-of-sample, and incorporate more realistic transaction-cost and capital-allocation models.